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Trends and amounts of consumption of low-calorie sweeteners: A cross-sectional study

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SUMMARY

Background and aims: White sugar has been blamed to cause health consequences including diabetes and obesity. With the increased need to decrease sugar intake, attention was shifted towards sugar alternatives also known as low-calorie sweeteners. Low-calorie sweeteners are ubiquitous within food products, but their consumption among Lebanese adults isn't well elucidated. Our study aims to investigate the trends and amounts of low-calorie sweeteners' consumption in addition to their main sources among a sample of Lebanese adults.

Methods: A cross-sectional study was conducted over 384 adult Lebanese individuals from both genders residing in Beirut and Mount-Lebanon from February to August 2020. After excluding eight individuals, 376 completed the survey. To assess the trends, frequency and quantity of low-calorie sweeteners' consumption, a questionnaire was filled by trained and licensed dietitians in addition to a previously validated food frequency questionnaire adapted to the most widely known products in the Lebanese market. Food products were divided into categories, amount of sweeteners in each item was obtained from the label, and analysis was performed according to the type of sweetener they are made of. Frequencies were converted into numbers of servings per day and multiplied by the standard portion size. Daily individual consumption of each sweetener apart was obtained by multiplying the amount of sweetener in each item consumed by total daily consumption. The amount in mg/kg was obtained by dividing total consumption of each sweetener apart by the mean weight of our population. The Statistical Package for Social Sciences (SPSS) was used for data entry and analysis.

Results: Results showed that 94.4% were consuming an artificially-sweetened item at least once in the last six months. While weight loss and healthy lifestyle constituted the main reasons for past and current consumption respectively, safety and unpleasant taste affected consumption trends. Pills and powders' consumption was significantly associated with gender ($p = 0.032$) and dieting ($p = 0.000$) but not with age ($p = 0.831$), educational level ($p = 0.294$), living district ($p = 0.421$) or income ($p = 0.828$) where women and dieters used significantly more pills and powders than their counterparts. "Food and beverages" consumption was independent of all factors except age ($p = 0.001$); it peaked at 26–40 and declined thereafter. Consumption of low-calorie sweeteners came mainly from beverages and was as follows while remaining within the acceptable daily intake limits: aspartame: 98.9 mg/d (1.38 mg/kg/d); sucralose: 22.19 mg/d (0.31 mg/kg/d); acesulfame-K: 39.12 mg/d (0.55 mg/kg/d); stevia 3.28 mg/d (0.05 mg/kg/d). Food and beverages were consumed more than pills and powders due to unintentional consumption.

Conclusion: Despite that safety and unpleasant taste affected consumption of low-calorie sweetened products, intake of such products was found to be highly prevalent in the studied population. While remaining within the acceptable daily intake limits, consumption of food and beverages exceeded that of pills and powders due to unintentional consumption making awareness a necessity to help consumers make an informed decision. Moreover, a nation-level study is needed to generalize the results.

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1. Introduction

Sugar, the "white demon", has been blamed for increasing the risk for diabetes, obesity and cardiovascular diseases [1] to cite few. This have pushed health authorities to recommend limiting intake of white sugar to less than 10% of total calories [2,3]. With the increased need to decrease sugar intake, attention was shifted towards sugar alternatives also known as high-intensity sweeteners, non-nutritive sweeteners (NNS), artificial sweeteners or low-calorie sweeteners (LCS), knowing that all those terms were used interchangeably in the literature to describe the alternatives to sugar.

Sugar alternatives first appeared in the 1800s [4] and thousands of products that contain sugar substitutes are introduced to the market yearly. While eleven are approved for use in the European Union including Acesulfame-K (ace-K) (E950), Aspartame (E951), Cyclamate (E952), Saccharin (E954), Sucralose (E955), Thaumatin (E957), Neohesperidine DC (E959), Steviol Glycosides (E960), Neotame (E961), Aspartame-Acesulfame Salt (E962) and Advantame (E969) [5], eight are approved for human consumption in the United States (US) including Saccharin, Aspartame, Ace-K, Sucralose, Neotame, Advantame, Steviol glycosides, and Luo Han Guo fruit extracts [6]. In the states, the first six are regulated by the Food and Drug Administration (FDA) as food additives and require premarket approval whereas the last two are generally recognized as safe and don't require FDA approval [7]. All are 30–13,000 times sweeter than sugar [8], enter in more than 6000 item in the US [9], and have different functional properties that may affect perceived taste or use in different food applications. They are ubiquitous within food products [10] such as dairy products, sweetener packets [11], desserts, gum, breakfast foods and diet beverages, and therefore may be unintentionally consumed [12].

Several studies have investigated the intake of sweeteners among different populations and ages while using different categories and combinations including LCS beverages, tabletop sweeteners, LCS food, LCS food and beverages (FB) … etc. where items included in each category differed among studies. Drewnowski and Rehm [13] who used data from five National Health and Nutrition Examination Survey's (NHANES) cycles from 1999 to 2008, found that 30% of US adults were consuming some type of LCS with 19% consuming LCS beverages, 11.4% consuming tabletop sweeteners, and 4.6% consuming LCS food. Another study used data from NHANES 2007–2012 and found that 47.8% of US adults reported using more than one item containing NNS over two days [14]. Sylvetsky et al. [15] used data from NHANES 2009–2012 and concluded that 41.4% of adults were using LCS. In Lebanon, estimation on dietary exposure to sweeteners is limited. For this reason, our main purpose was to investigate the trends and amounts of consumption of LCS among a sample of Lebanese adults. Outcomes will indicate the trends in consumption, the main sources of LCS in addition to the amount consumed.

2. Materials and methods

2.1. Study design and sample size calculation

A cross sectional study was conducted over 384 adult Lebanese individuals from both genders from February to August 2020. The sample size (n) was based on data published by the Lebanese Ministry of Public Health in the statistical bulletin-2018 [16], and was calculated following the Krejcie-Morgan formula [17]: $n = \frac{X^2xNxP(1-P)}{e^2(N-1)+X^2xP(1-P)} = 384$ where X^2: chi-square of degree of freedom 1 and confidence 95% = 3.841; N: population size = 1,374,082; P: proportion of population = 0.5, and e: acceptable sampling error = 0.05. The percentage of each district in the general population was similarly represented in the study sample. After excluding eight individuals for not completing the questionnaire, a total of 376 participants was retained.

2.2. Data collection

Participants were recruited from Beirut and Mount-Lebanon. Trained and licensed dietitians filled the questionnaires with the participants after explaining the study and obtaining their oral consent for participation. Seeing that each district is composed out of several villages and cities located at different altitudes spread between zero and 1320 m over sea level, we made sure to include in our sample individuals living at different altitudes due to the difference in products' availability and lifestyle. In addition, we made sure not to include more than one person from the same household.

2.3. Measurement tools

To assess the trends, frequency and quantity of LCS consumption, we designed a questionnaire made up of five parts covering personal and socioeconomic status, educational level, daily habits, LCS consumption and knowledge, anthropometric measurements (height and weight) in addition to the food frequency questionnaire (FFQ). Due to the lack of a validated FFQ assessing LCS consumption over Lebanese food, a market study in three big supermarkets located in our research area was conducted between August and September 2019. The list was updated in January 2020. A previously validated questionnaire [18] was used and adapted to the most widely known products in the Lebanese market. Food products were divided into categories, amount of sweeteners in each item was obtained from the label, and analysis was performed according to the type of sweetener they are made of. Frequencies were converted into numbers of servings per day and multiplied by the standard portion size. Daily individual consumption of each sweetener apart was obtained by multiplying the amount of sweetener in each item consumed by total daily consumption. The amount in mg/kg was obtained by dividing total consumption of each sweetener apart by the mean weight of our population. Weight and height were self-reported.

2.4. Statistical analysis

The Statistical Package for Social Sciences (SPSS) version 22.0 was used for data entry and analysis. The confidence interval (CI) was set at 95%, and significance was considered at p-value of <0.05. Descriptive statistics were performed on data about personal and socioeconomic status, educational level, daily habits, LCS consumption and knowledge, and anthropometrics. Results were presented as percentages for qualitative variables, and as mean and standard deviation (SD) for quantitative variables. Cross-tab and X^2 were used to study the association between different qualitative variables including consumption of LCS pills and powders (pp) and LCS FB, risk perception, age, gender, educational level, dieting, living district and income.

2.5. Ethical statement

The study was conducted according to the guidelines of the Declaration of Helsinki, and approved by the Institutional Review Board of CENTRE HOSPITALIER UNIVERSITAIRE- NOTRE DAME DES SECOURS (December 20, 2019). Oral consent was obtained from subjects prior to participation. Questionnaires were anonymous; all data collected was kept confidential and was used for academic purposes only.

3. Results

3.1. Description of the population's general characteristics

A total of 376 participants (137 males and 239 females) living in Beirut and Mount-Lebanon (21.5% and 78.5% respectively) were included in the study. The mean age of our participants was 31.89 ± 10.44 years, and their mean body mass index (BMI) was 24.97 ± 4.49kg/m2. The majority of our population was single (72.9%; n = 274), holding a university degree (72.6%; n = 273), and had an income ≥1.500.000 L.L per month (58.8%; n = 143). The majority of our subjects had a sedentary lifestyle (39.9%; n = 150), and is not currently following a weight loss diet (65.2%; n = 245). More characteristics of our population can be found in Table S1.

3.2. Lifetime consumption of low-calorie sweeteners

Sweeteners can be found in pills and powders or within food items. In our study, 66.8% (n = 251) of our population have never used LCS pp before, while 33.2% (n = 125) have added them to their sweets or drinks at least once in their life. When it comes to food and beverages containing LCS, 95.5% (n = 359) have consumed them at least once versus 4.5% (n = 17) who have never tried them before.

3.3. Past and current consumption of low-calorie sweeteners

To assess the trends in consumption, we divided it into past and current/recent consumption where current consumption refers to the consumption of LCS food items (including two categories: "food and beverages", and "pills and powders") in the last six months, whereas past consumption refers to the consumption of such products before that period.

3.3.1. Consumption and reasons for consumption of products containing low-calorie sweeteners

Only 29.3% (n = 110) of our participants reported using pills and powders in the past, whereas 83.2% (n = 313) were using food and beverages (Fig. 1) mainly for weight loss (Fig. 2). The same trend was reported for recent consumption where 20.5% (n = 77) reported current usage of pp versus 94.1% (n = 354) for LCS food and beverages (Fig. 1).

The healthy lifestyle low in calories and white sugar constituted the main reason for current consumption of LCS items (29.6%; n = 98), followed by a combination of causes (mainly including healthy lifestyle, weight loss, and favorable taste among others). Notably, 14.5% of our population (n = 48) were not aware that the product they are consuming contains a LCS; this percentage increased to 18.4% when we considered the individuals who reported being unaware within the "two causes or more" category. Other reasons are shown in Table 1.

3.3.2. Reasons to stop past consumption of products containing low-calorie sweeteners

Among those who were consuming pp (n = 110) and FB (n = 313) containing LCS in the past, 44.5% (n = 49) have stopped consuming pills and powders, and 3.5% (n = 11) have stopped consuming LCS food and beverages (Table 2) mainly for safety concerns (26.5%, n = 13 and 72.7%, n = 8 respectively) (Table 3). 24.5% (n = 12) had other reason to stop using pp (mainly because they prefer to have their drinks unsweetened), while 20.4% (n = 10) have stopped using them because of their unpleasant taste. Among past FB consumers, 18.2% (n = 2) have stopped using them for a combination of causes, and 9.1% (n = 1) because they didn't like their taste (Table 3).

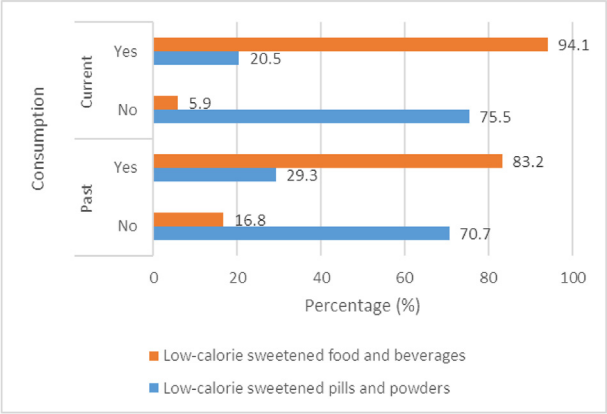


Fig. 1. Past and current consumption of LCS food and beverages, and LCS pills and powders.

3.3.3. Reasons behind the lack of current consumption of items containing low-calorie sweeteners

Among current non-consumers, the main reason for not using pp is the fact that subjects had other reasons (28.8%; n = 85) for not using them (mainly because they prefer to drink their hot drinks without any sweetener added). When it comes to the remaining non-users, 27.1% (n = 80) had safety concerns, 24.1% (n = 71) found their taste unpleasant, 14.2% (n = 42) believed that there is no difference between LCS pp and white sugar, and 2.4% (n = 7) found them expensive. Although only 5.9% (n = 22) of our total population was not consuming LCS food and beverages in the last six months (Table 4), the majority was not taking them for the same reasons of safety (52.4%; n = 11) and taste (23.8%; n = 5) (Table 5).

3.4. Association between current consumption of pp and FB containing low-calorie sweeteners and different variables (including age, gender, educational level, weight loss, living district and income)

Pills and powders' consumption was found to be significantly associated with gender (p = 0.032) and dieting (p = 0.000) but not with age (p = 0.831), educational level (p = 0.294), living district (p = 0.421) or income (p = 0.828) (Table S2).

When it comes to the consumption of FB containing LCS, a significant relationship was found with age (p = 0.001) but not with gender (p = 0.994), educational level (0.597), dieting (p = 0.089), living district (p = 0.282) or income (p = 0.131) (Table S3).

3.5. Association between risk perception of low-calorie sweeteners and different variables (including consumption of food and beverages, pills and powders, age, gender, educational level and dieting)

Risk perception of LCS is represented in Figure S1 where 62.8% of our participants considered them as being harmful. Risk perception significantly influenced consumption where those who identified LCS as harmful were less likely to add them to their hot drinks (p = 0.009). Risk perception was also significantly associated with gender (p = 0.000) and educational level (p = 0.032) where the percentage of females and participants holding higher degrees who perceived LCS as harmful was significantly higher than their counterparts (Table S4).

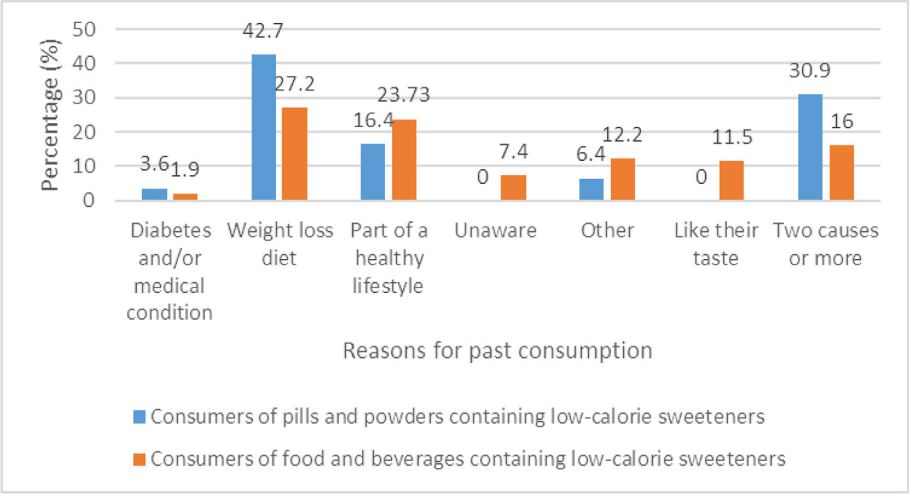


Fig. 2. Reasons for past consumption of LCS food and beverages, and LCS pills and powders.

Table 1
Reasons for current consumption of LCS items (from food and beverages, and pills and powders).

Reasons								
	Diabetes and/or medical condition	Weight loss diet	Part of a healthy lifestyle	Like their taste	Unaware	Two causes or more	Other	Total
Percent (%)	2.1%	13%	29.6%	13.3%	14.5%	15.1%	12.4%	100%
Frequency	7	43	98	44	48	50[a]	41	331

[a] Among them, 13 individuals were unaware that the product they are consuming contains a low-calorie sweetener.

Table 2
Stopped consumption of items containing LCS.

Stopped consumption			
Items containing LCS	No	Yes	Total
Pills and powders	61 (55.5%)	49 (44.5%)	110 (100%)
Food and beverages	302 (96.5%)	11 (3.5%)	313 (100%)

Values represent number of respondents (percentage); abbreviations: LCS: low-calorie sweeteners.

Table 3
Reasons to stop consumption of "pills and powders", and "food and beverages" containing LCS.

Reasons							
Items containing LCS	Harmful	Expensive	Taste dislike	No difference between LCS and white sugar	Other	Two causes or more	Total
Pills and powders	13 (26.5%)	2 (4.1%)	10 (20.4%)	7 (14.3%)	12 (24.5%)	5 (10.2%)	49 (100%)
Food and beverages	8 (72.7%)	0 (0%)	1 (9.1%)	0 (0%)	0 (0%)	2 (18.2%)	11 (100%)

Values represent number of respondents (percentage).

Table 4
Current consumption of "pills and powders", and "food and beverages" containing LCS.

Current consumption			
Items containing LCS	No	Yes	Total
Pills and powders	299 (79.5%)	77 (20.5%)	376 (100%)
Food and beverages	22 (5.9%)	354 (94.1%)	376 (100%)

Values represent number of respondents (percentage).

3.6. Trends in current consumption of low-calorie sweeteners

It has been shown that LCS were consumed as follows: 98.9 mg (1.38 mg/kg) of aspartame, 22.19 mg (0.31 mg/kg) of sucralose, 39.12 mg (0.55 mg/kg) of ace-K, and 3.28 mg (0.05 mg/kg) of stevia per day (Table S5). While no one exceeded the acceptable daily intake (ADI) for aspartame or ace-K, four individuals in the sucralose group and one individual in the stevia group exceeded the ADI limits with a total consumption of 164.2%, 144.5%, 107.9%, 147%, and 176.1% of the ADI for sucralose and stevia respectively (data not shown).

Our present study shows that 94.4% (n = 355) of our population have consumed at least one item containing LCS (whether from pp

Table 5
Reasons of current non-consumption of "pills and powders", and "food and beverages" containing LCS.

Reasons							
Items containing LCS	Harmful	Expensive	Taste dislike	No difference between LCS and white sugar	Other	Two causes or more	Total
Pills and powders	80 (27.1%)	7 (2.4%)	71 (24.1%)	42 (14.2%)	85 (28.8%)	10 (3.4%)	295 (100%)
Food and beverages	11 (52.4%)	1 (4.8%)	5 (23.8%)	2 (9.5%)	2 (9.5%)	0 (0%)	21 (100%)

Values represent number of respondents (percentage).

or FB) in the last 6 months (Figure S2). When divided by category, food constituted the most monthly consumed LCS item where 82.4% (n = 310) of people reported consuming a LCS food at least once per month. Beverages came in the second place where 71.8% (n = 270) of our population have consumed an artificially sweetened beverage (ASB) at least once per month. The least item consumed was pills and powders where 18.1% (n = 68) of our subjects reported taking LCS pp at least once per month. The same trend was reported for daily consumption with LCS food items being on the top of the list (36.7%; n = 138), followed by ASB (22.3%; n = 84) and pp (9%; n = 34) (Table S6). Gums and candies and diet sodas constituted the most items consumed within the food (Table S7) and beverages categories (Table S8). But when we assessed the source of sweeteners consumed, we found that 78% of aspartame, 54.9% of sucralose, 81.4% of ace-K and 79.9% of stevia consumed per day came from beverages (Table S5).

4. Discussion

Our present study shows clearly that the consumption of FB containing LCS exceeded by far pp consumption. Even though the majority of the population under study was not consuming pp, women were more likely to use them than males, as were those who followed a weight loss diet. This is consistent with several studies [11,13,14] and can be reported to the fact that females tend to be more concerned about their weight and caloric intake than males, and people who are dieting use LCS instead of sugar for caloric control. When it comes to LCS food and beverages, consumption was independent of all factors (including gender, educational level, dieting, living district or income) except age; it peaked at 26–40 years and declined thereafter which is inconsistent with several other studies [13,19] where consumption was found to be most common among those aged 55–74 years [11]. This inconsistency can be reported to the fact that our population is relatively young and have a mean age of 31.89 ± 10.44 years, and the number of older adults is relatively low which might have affected the results. However, other factors didn't affect the consumption of low-calorie sweetened FB most probably because LCS are ubiquitous within food products, and a large group of people are taking them while being unaware that they contain LCS. Consistently with the available literature [20–22], unawareness, lack of knowledge, and safety concerns were well documented in our study where 14.5% of current consumers were unaware that the product they are consuming contains LCS, and half of the subjects thought that diet soda or sugar-free products contain white sugar or ignored that such products don't contain white sugar (data not shown). Only 14.9% believed that those products are not harmful (Figure S1) and 10.1% believed that they do not cause cancer (data not shown).

When divided by item, foods containing LCS were the most consumed, followed by beverages and pp. Despite that, ASB were found to contribute to more than 50% of the total daily amount of sweeteners especially that a single beverage contains significantly higher amount of sweeteners than any food or pp. This agrees with several studies that reported beverages to constitute the main source of LCS [11,14,20].

Safety and unpleasant taste were the most two common reasons for our participants to stop or not using pp or FB that contain LCS. Those two factors have constituted a debatable topic among scientists where health effects of LCS consumption are still inconclusive [23]. People found in artificial sweeteners the "decent" alternative to white sugar that provides a sweet taste without affecting caloric intake or glycemic response. However, this experience was not fully pleasant; bitterness and metallic off-flavors are one of the most common problems of LCS [24], and the unpleasant taste influenced product acceptability and was among the most common reasons driving our participants to stop LCS usage. Tan et al. [25] compared the side taste profiles of artificial sweeteners to sugar; they concluded that ace-K, aspartame, stevia and sucralose had significantly higher bitterness, chemical and metallic taste citations than sucrose. This could in part explain our results where all pp are made of one or more of the aforementioned LCS.

When it comes to safety, risk perception significantly influenced pp consumption among our subjects, and was associated with gender and educational level. This was consistent with a recent study where risk perception was significantly linked to a decreased consumption of LCS, and was significantly affected by gender and educational level especially among females and those holding higher degrees [20]. The literature investigating the safety of LCS is highly heterogeneous, and results vary with different studies' type and design among other factors. While some studies blamed LCS for contributing to diabetes [26–29], obesity [30–32], cardiovascular problems [33] and cancer [34], others proved their "innocence" [35–38]. In the midst of this contradiction, the American Diabetes Association claimed a potential benefit in consuming LCS instead of sugar, but encouraged people to decrease their intake of artificial sweeteners while emphasizing on the importance of water consumption [39], beside the FDA [7] and the Academy of Nutrition and Dietetics association [40] who claimed the safety of artificial sweeteners if used within the ADI.

According to Martyn et al. [41] "there are no concerns with respect to exceedance of individual sweetener ADIs among the general population globally". Consistently with this claim, no one in our population exceeded the ADI for aspartame or ace-K but 1.06% and 0.26% exceeded the ADI for sucralose and stevia respectively. As previously discussed, the exact effects of LCS on health are still unknown. For this reason and even though the percentage is very low, educating individuals about the risks associated with over-consumption of LCS is essential, and conducting a national study to assess the intake among all Lebanese is of utmost importance.

To our knowledge, this study was the first to investigate the trends and amounts of consumption of LCS among a sample of Lebanese adults. In our study, self-report method was not used because it constitutes one of the limitations in estimating LCS consumption [11]; instead, trained and licensed dietitians filled the FFQ with respondents to eliminate this bias. However, this study was not without limitations. Despite that we tried to cover the most common and well known products consumed by Lebanese, less common products were not included, and certain common products were removed (including some juices, energy drinks, chocolates, biscuits, candies, popsicles, non-dairy creamers, and some

artificially sweetened pills and powders) because their label lacked the amount of sweeteners which was not also available in the literature. For diet sodas, because the label was lacking the amount of sweetener(s), and because companies were not helpful in providing us the exact amount of sweetener(s) present, an estimate of sweeteners' amount was used based on their website [42] and on the literature [43]. Moreover, weight and height were self-reported; this might affect the amount in mg/kg but not the total amount of sweeteners that is independent of weight.

5. Conclusion

In conclusion, our study shows that despite that safety and unpleasant taste affected consumption of low-calorie sweetened products, intake of such products was found to be highly prevalent in the studied population. While remaining within the ADI limits, consumption of FB exceeded that of pp mainly because LCS are ubiquitous within food products and people tend to consume them unintentionally while being unaware that the product they are consuming contains LCS. Here comes the necessity of providing consumers with scientific information about sugar alternatives, educating them about the risks associated with overconsumption, and informing them on the importance of label checking to help them make an informed decision. Moreover, a nation-level study is needed to investigate the exact consumption among the Lebanese population which will generalize the results.

Statement of authorship

Mira Daher: Conceptualization, methodology, software, formal analysis, investigation, resources, data curation, writing—original draft preparation, writing—review and editing, visualization, project administration; Carla Fahd: investigation, resources; Afif Abdel Nour: validation, writing—review and editing, supervision; Yonna Sacre: Conceptualization, methodology, validation, writing—review and editing, supervision. All authors have read and agreed to the published version of the manuscript.

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Declaration of competing interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.clnesp.2022.01.006.

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